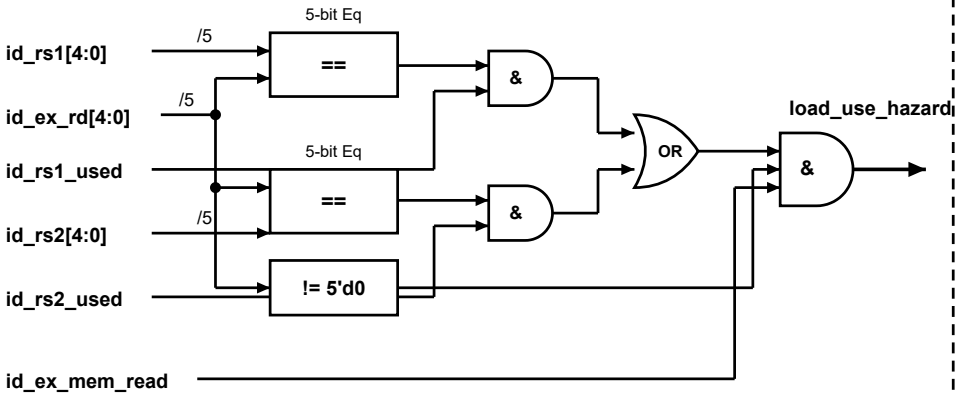
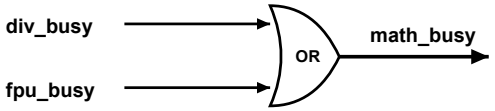
HAZARD DETECTION & DATA FORWARDING UNIT — HARDWARE SCHEMATIC

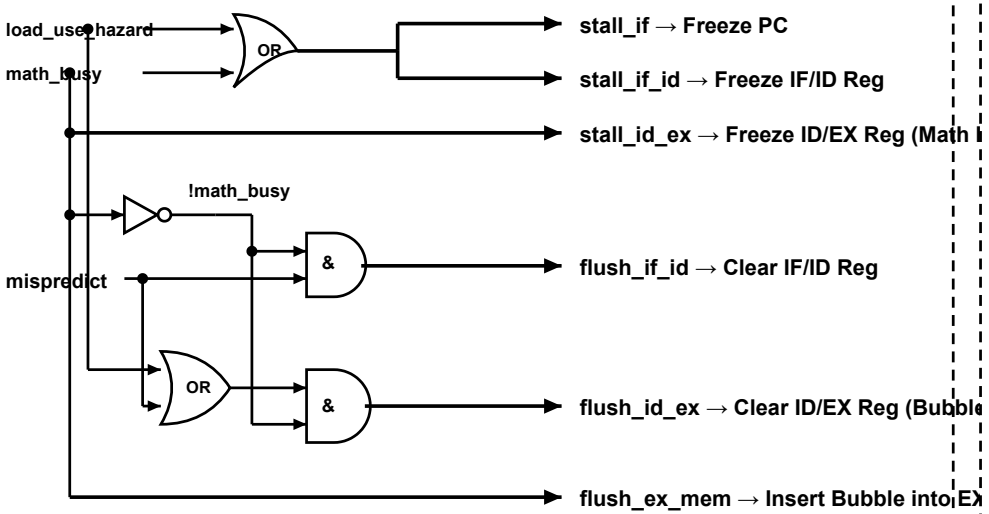
RV32IMF 5-Stage Pipelined Processor Core (Feature 7 / Feature 9 Architecture)

1. LOAD-USE & MULTI-CYCLE MATH INTERLOCK DETECTION



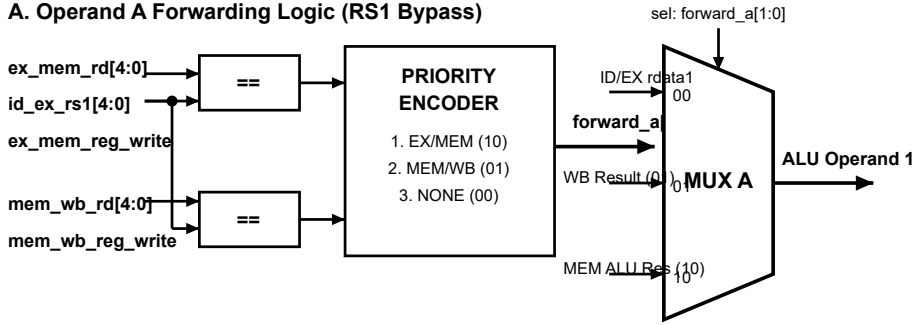


2. PIPELINE STALL & FLUSH GENERATION LOGIC

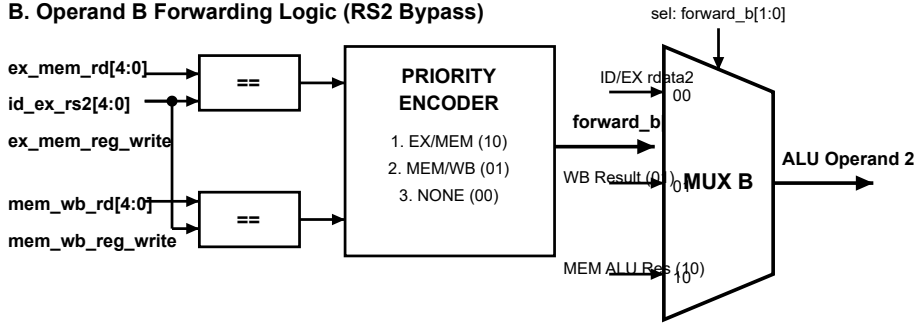


3. DATA FORWARDING UNIT (OPERAND A & OPERAND B)

A. Operand A Forwarding Logic (RS1 Bypass)



B. Operand B Forwarding Logic (RS2 Bypass)



BRANCH TARGET BUFFER (BTB) & SPECULATIVE RESOLUTION ARCHITECTURE

Cycle-Accurate Hardware Block Diagram • Query Path (IF), Pipeline Propagation, and EX-Stage Resolution

